

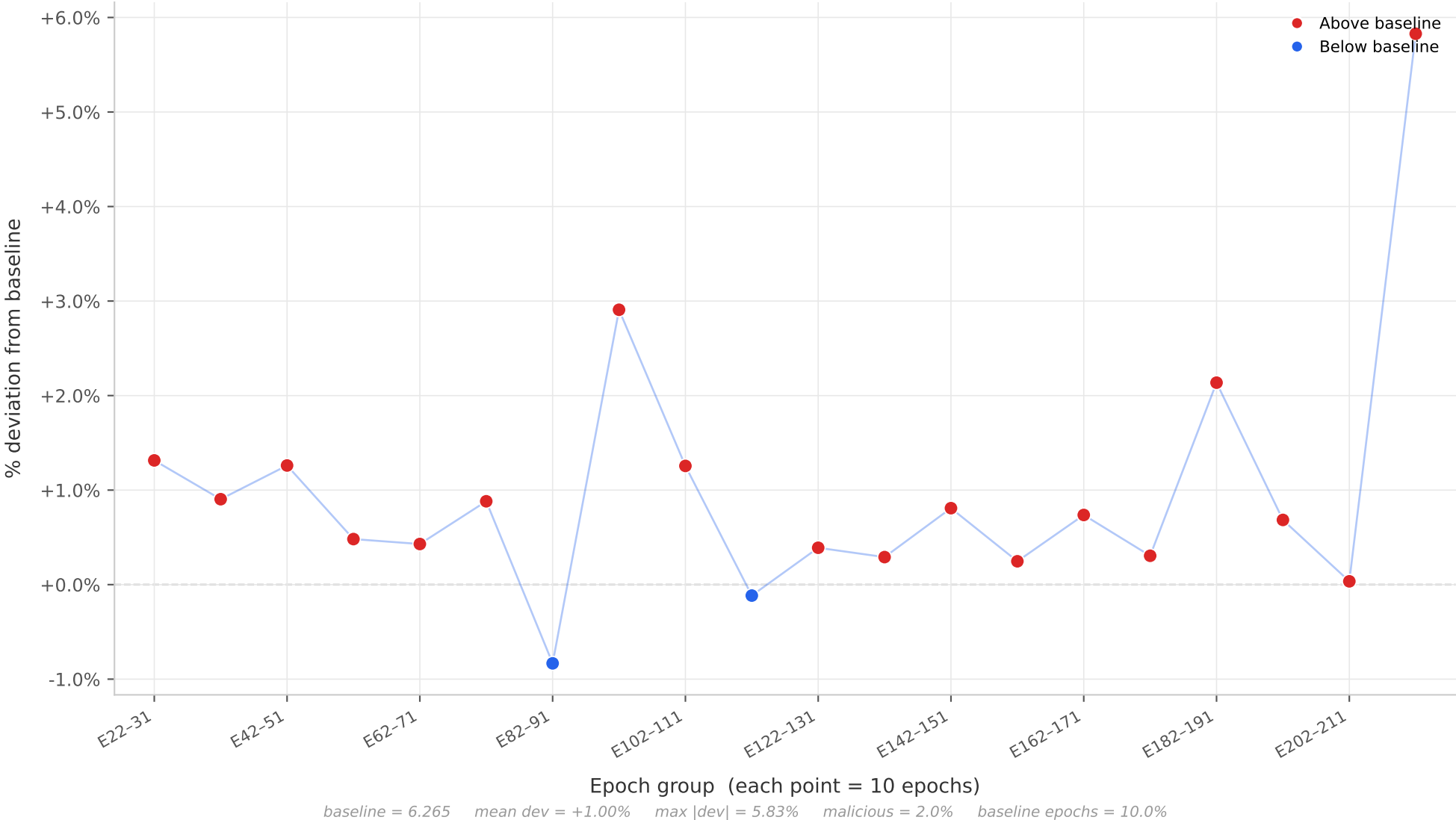
Entropy Stream Analysis

Deviation from Baseline — Per-Metric Report

Run Parameters		Baseline Values	
Dataset	epochs_10.0_2.0.json	Src IP Entropy	6.265
Directory	CAIDA_Dataset/txtD280/epochs/	Dst IP Entropy	6.066
Timestamp	2026-02-19T18:44:29.717503	Src Port Entropy	2.343
Total epochs	215	Dst Port Entropy	6.237
Baseline epochs	21 (10.0%)	Src IP Cardinality	5045
Attack epochs	194	Dst IP Cardinality	5027
Malicious traffic	2.0%	Src Port Cardinality	4302
Epoch group size	10 epochs per point	Dst Port Cardinality	5042
Metrics plotted	16	Src IP Uniformity	0.01534
Mode	FULL	Dst IP Uniformity	0.01337
		Src/Dst Ratio	1.003
		Total Flows	5078
		Unique Src IPs	3579
		Unique Dst IPs	2804
		Unique Src Ports	2323
		Unique Dst Ports	3825

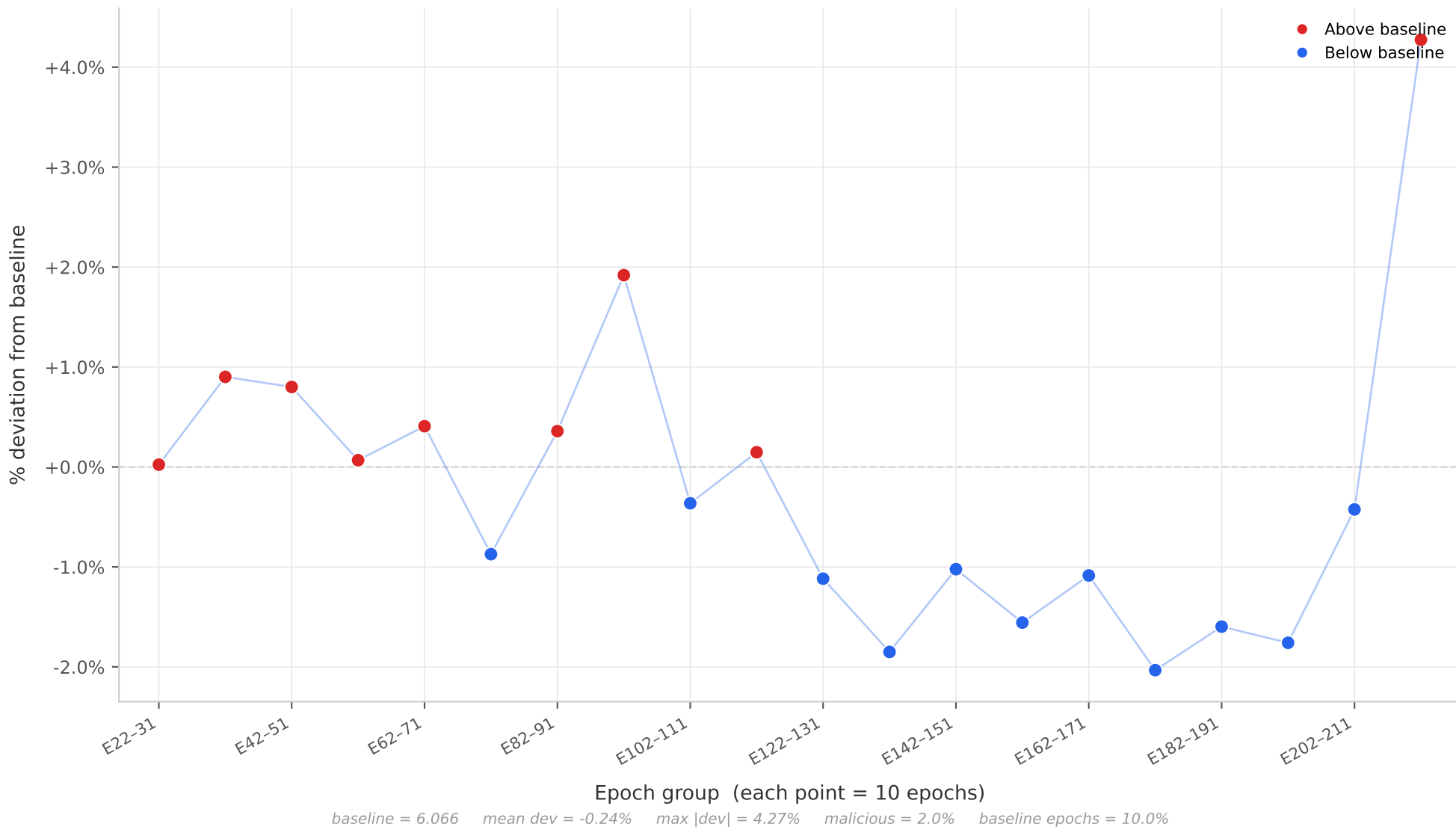
Source IP Entropy

Entropy



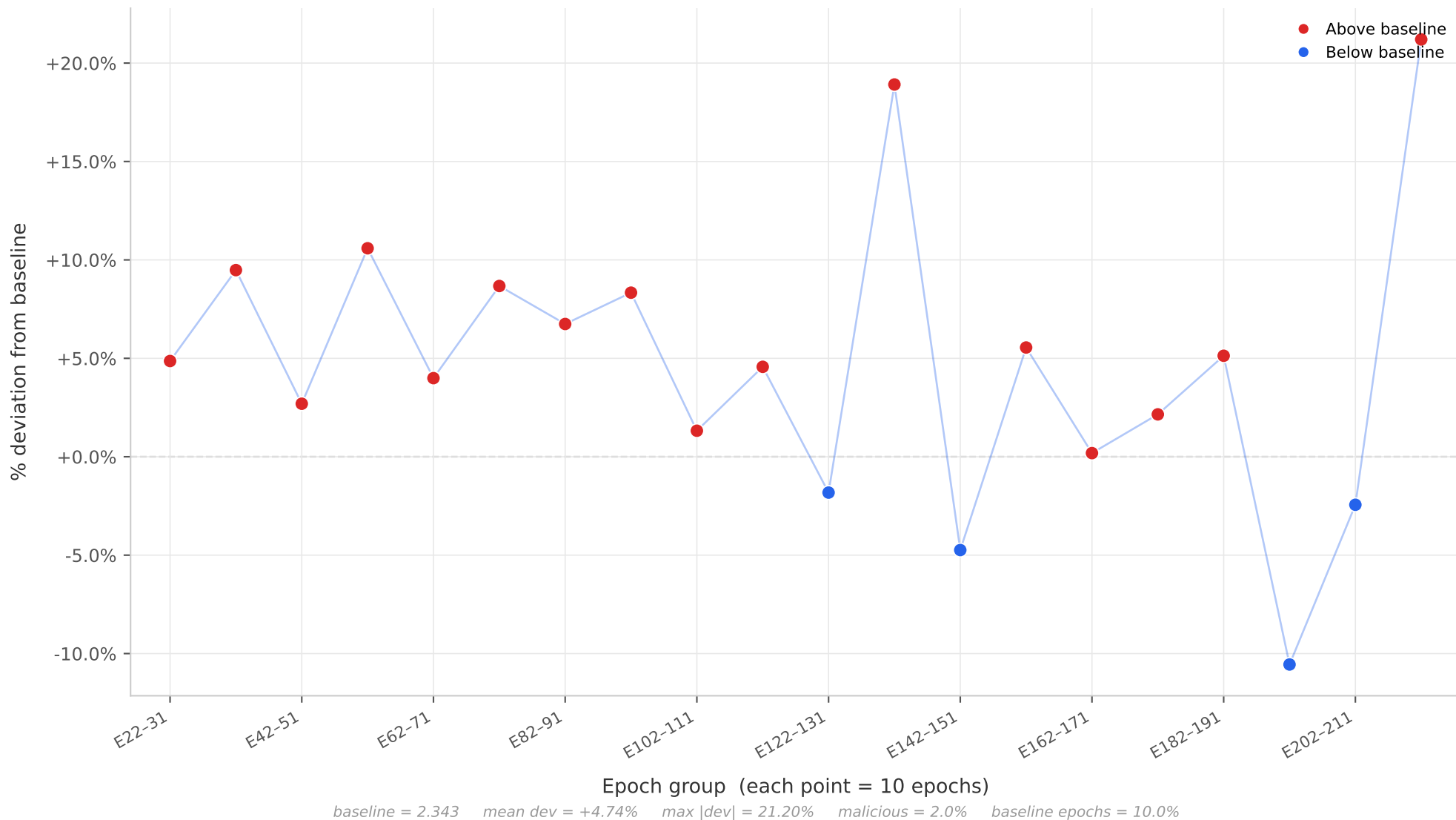
Destination IP Entropy

Entropy



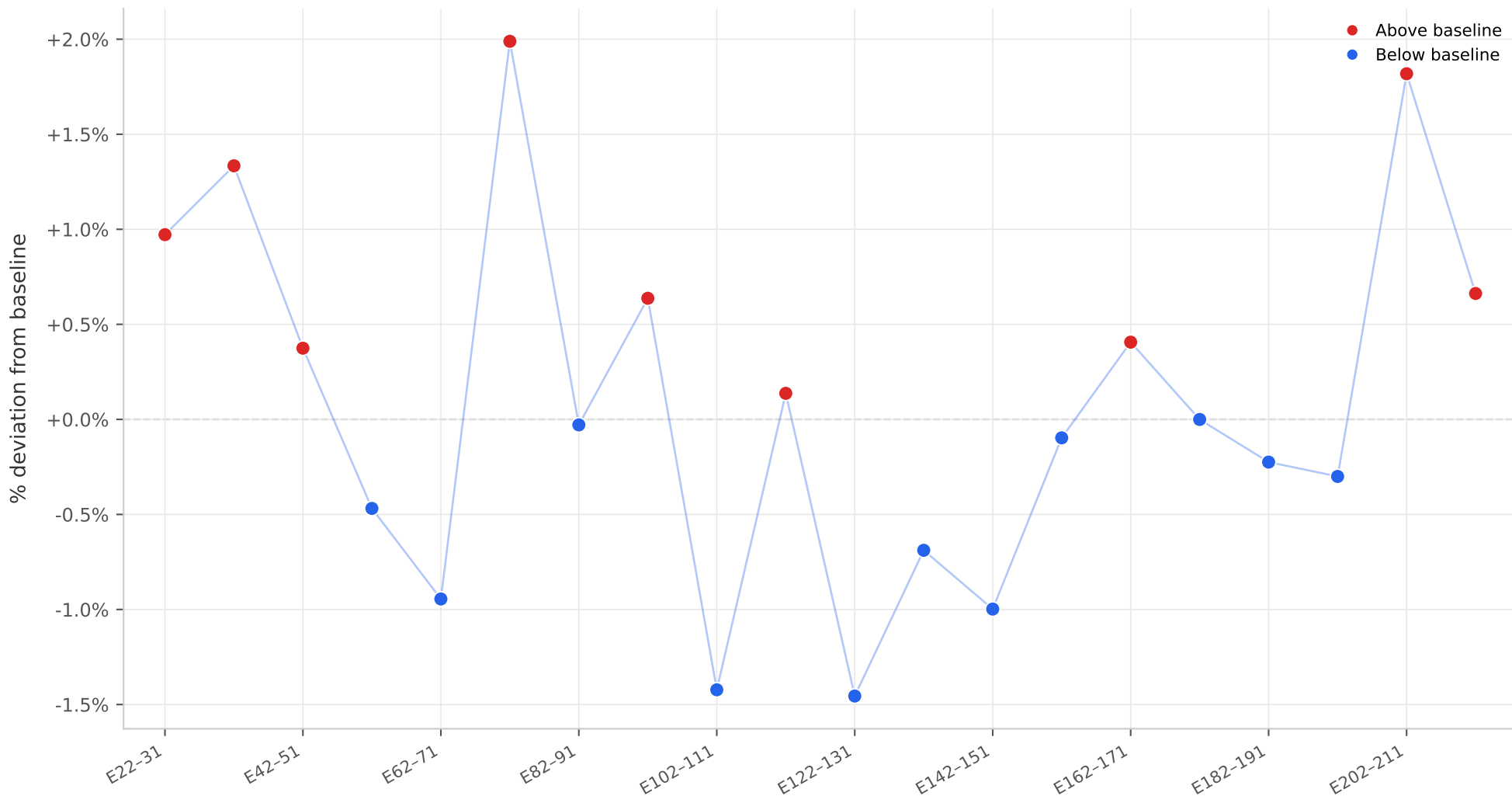
Source Port Entropy

Entropy



Destination Port Entropy

Entropy

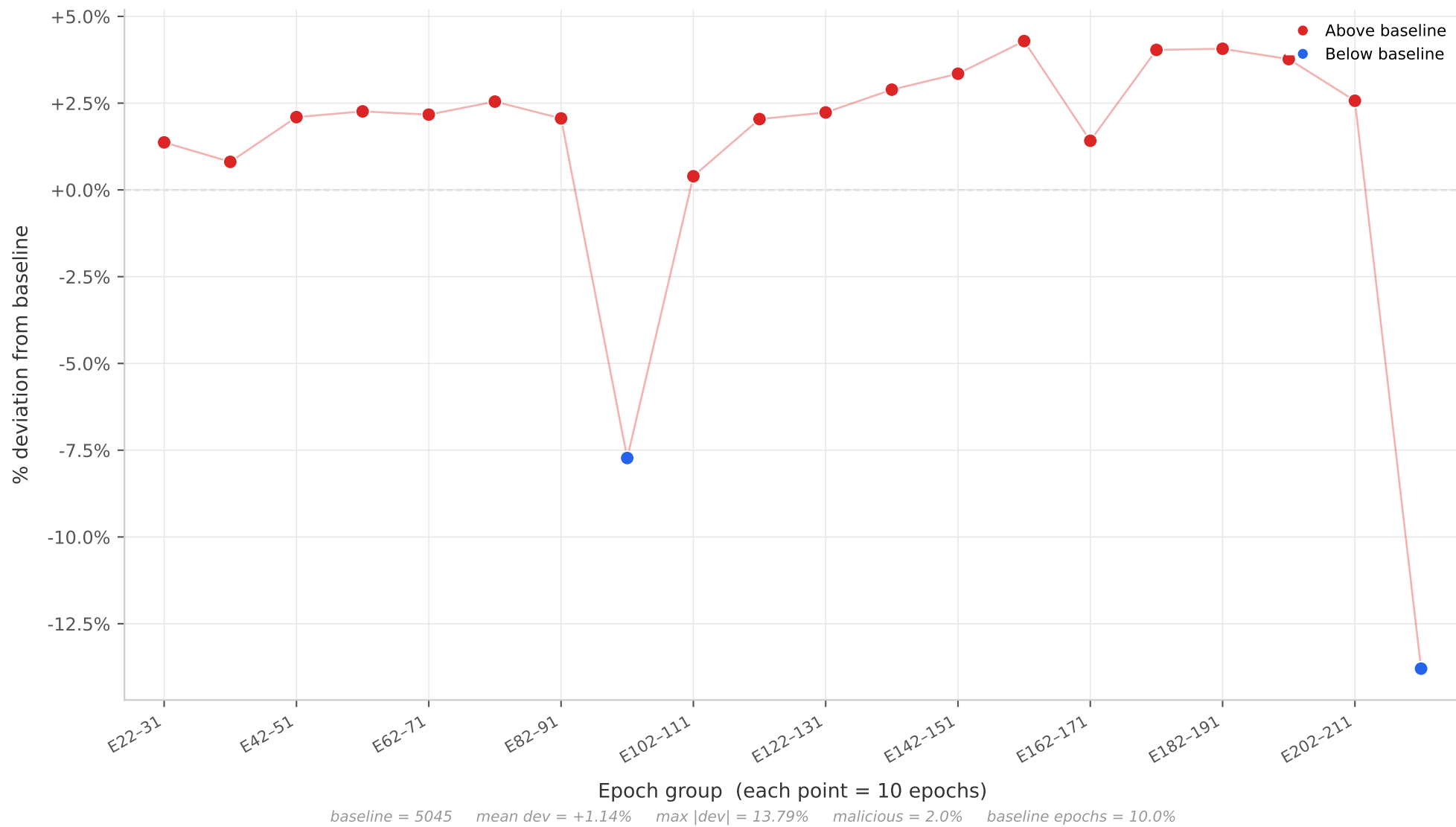


Epoch group (each point = 10 epochs)

baseline = 6.237 mean dev = +0.09% max |dev| = 1.99% malicious = 2.0% baseline epochs = 10.0%

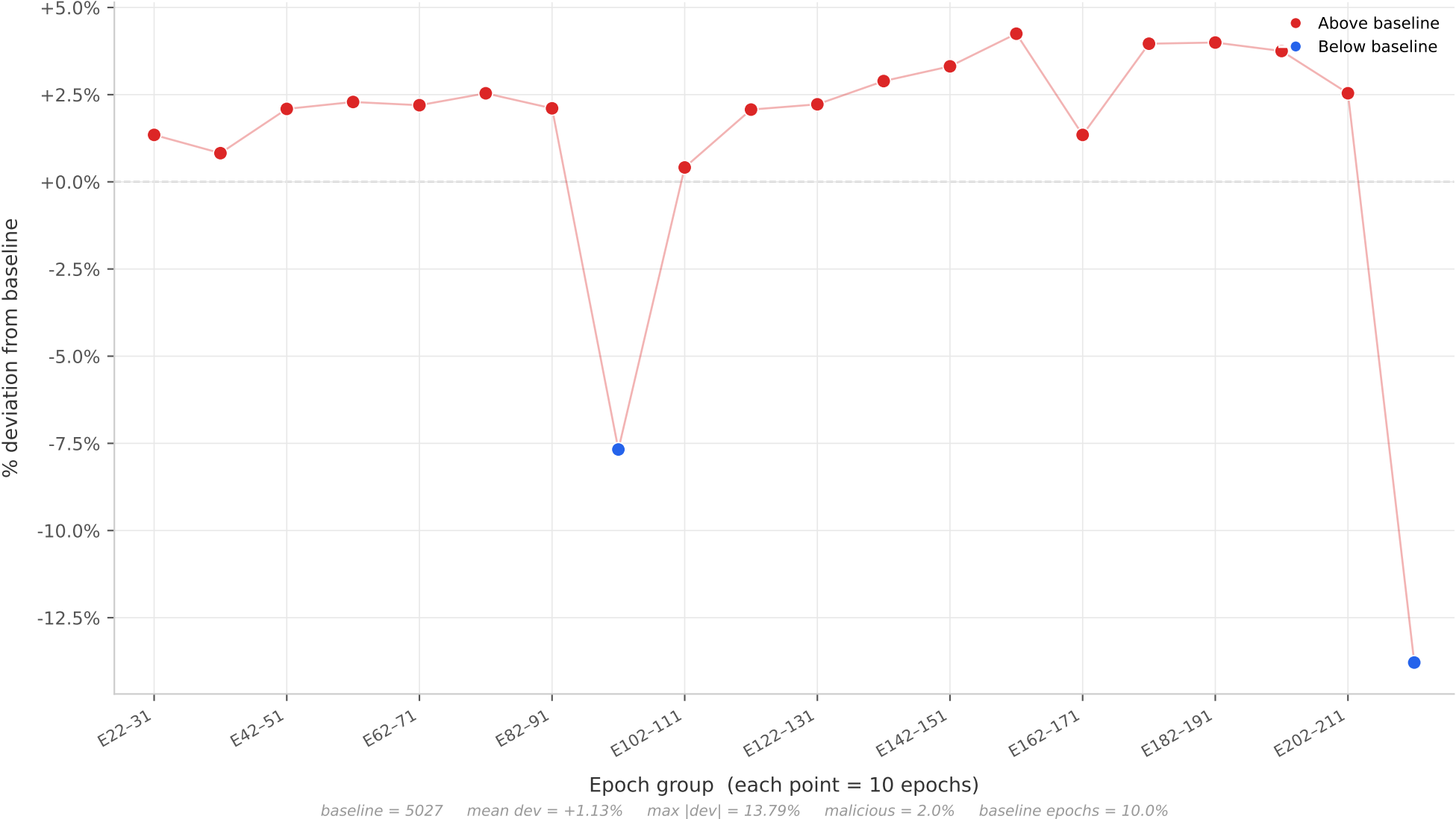
Source IP Cardinality

Cardinality



Destination IP Cardinality

Cardinality



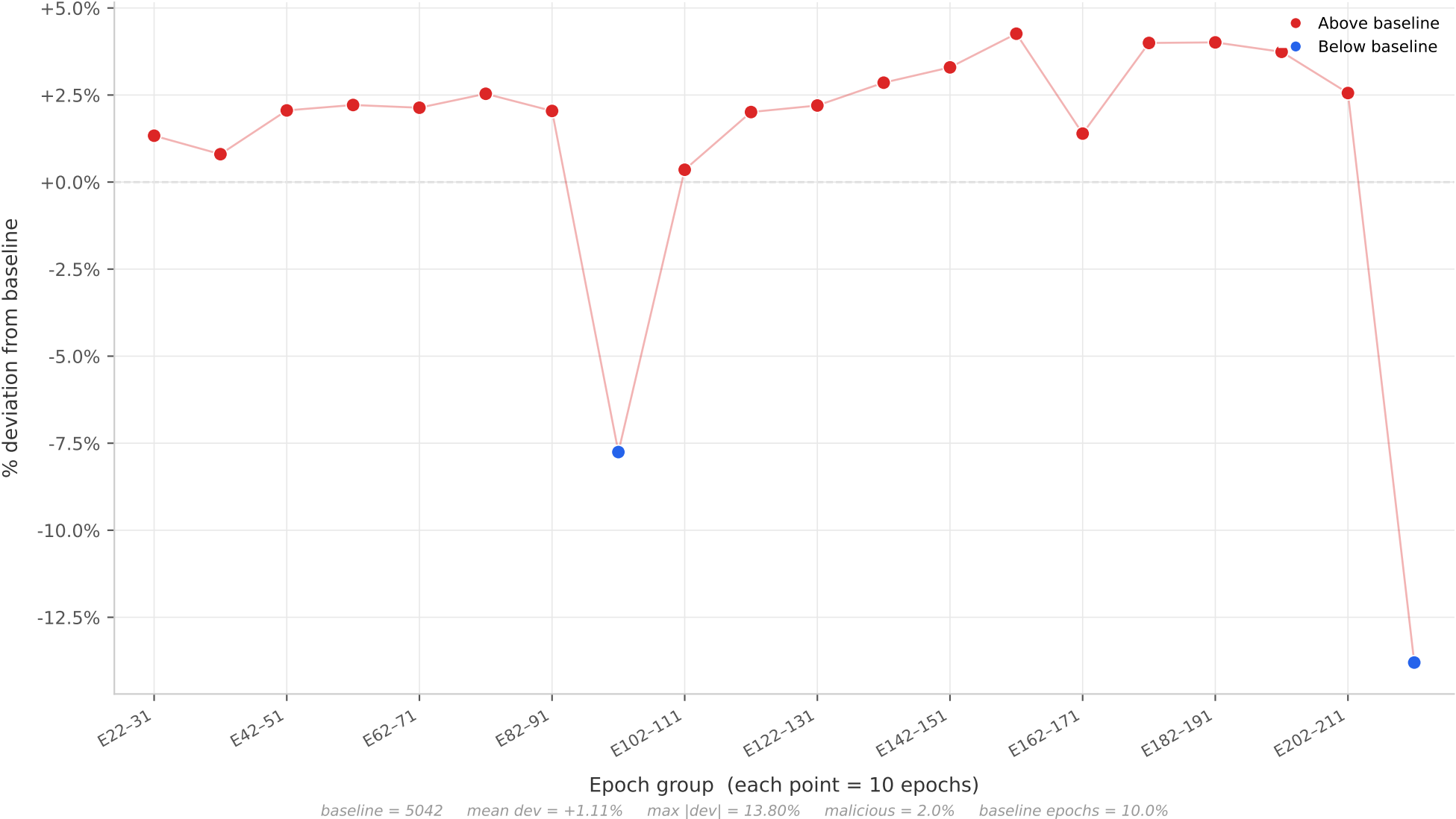
Source Port Cardinality

Cardinality



Destination Port Cardinality

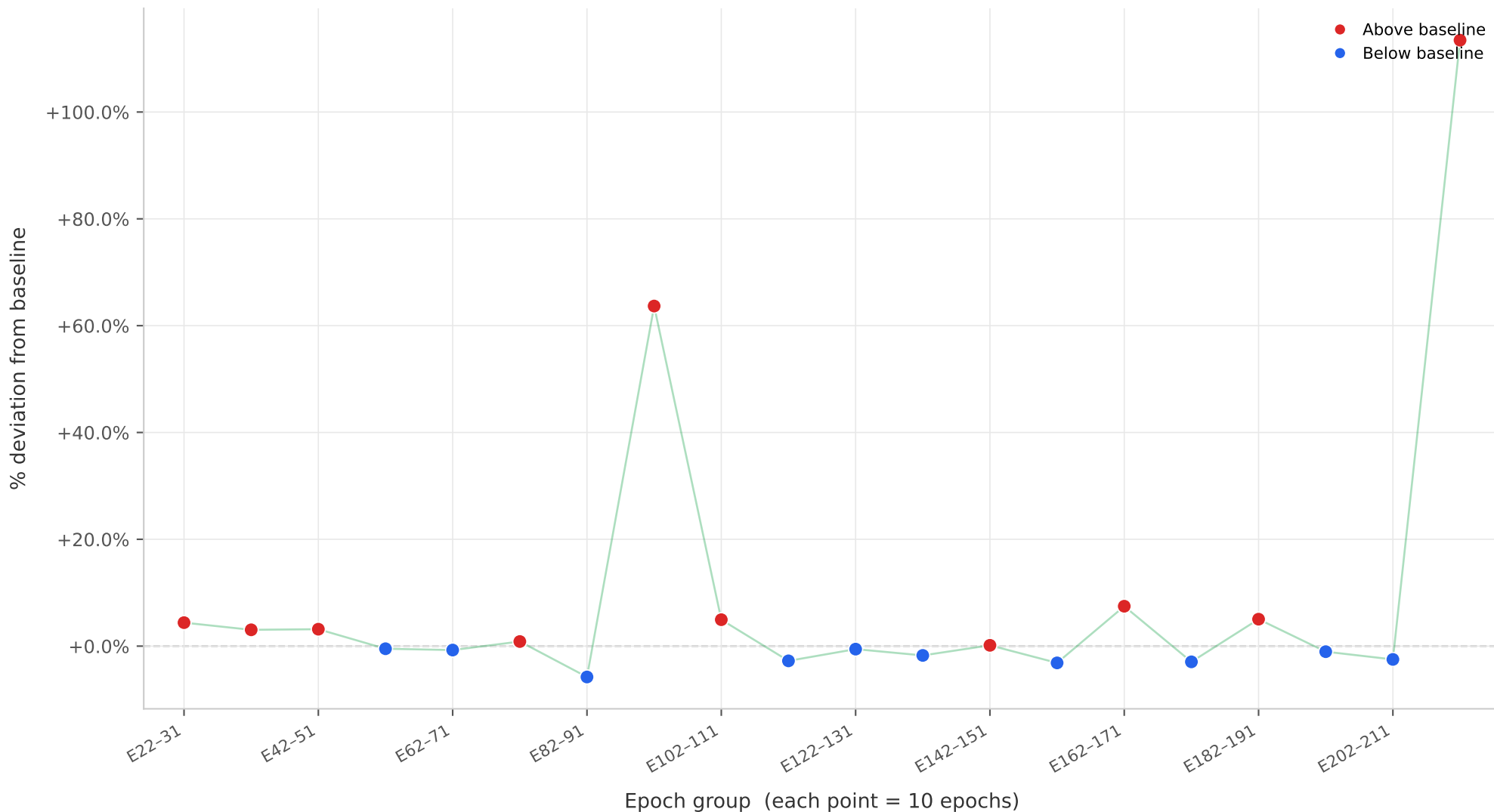
Cardinality



Source IP Uniformity

Uniformity

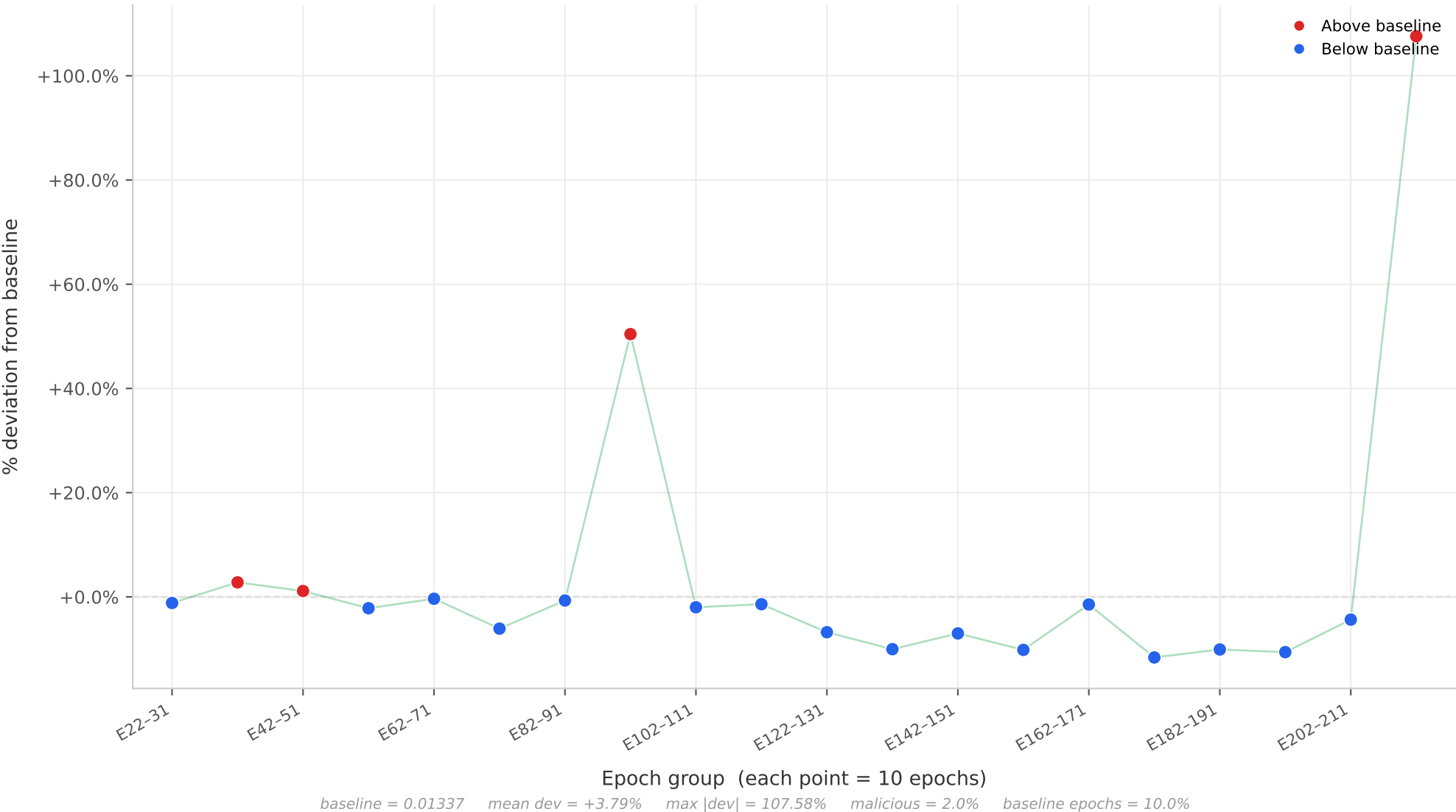
- Above baseline
- Below baseline



baseline = 0.01534 mean dev = +9.23% max |dev| = 113.45% malicious = 2.0% baseline epochs = 10.0%

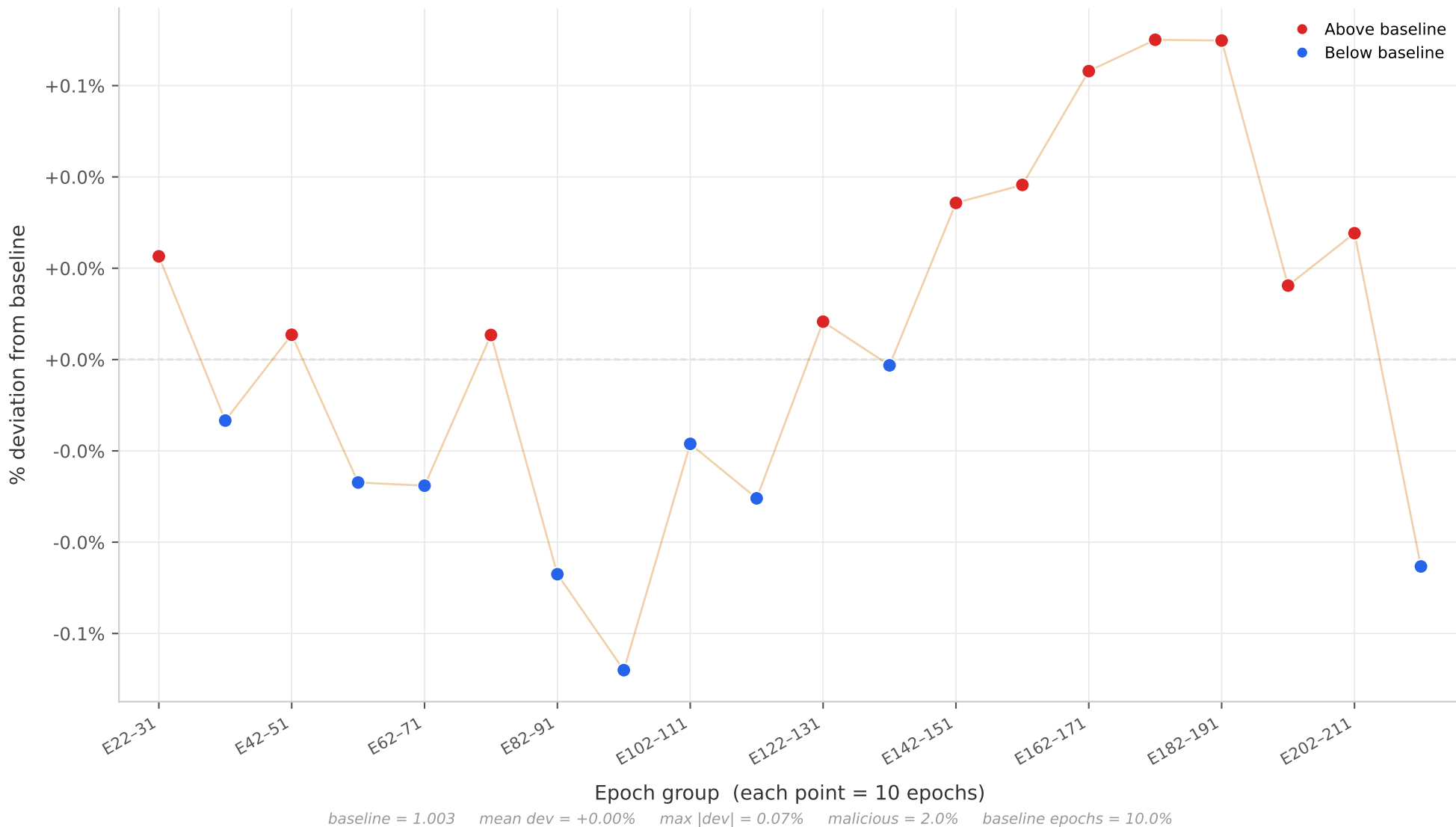
Destination IP Uniformity

Uniformity



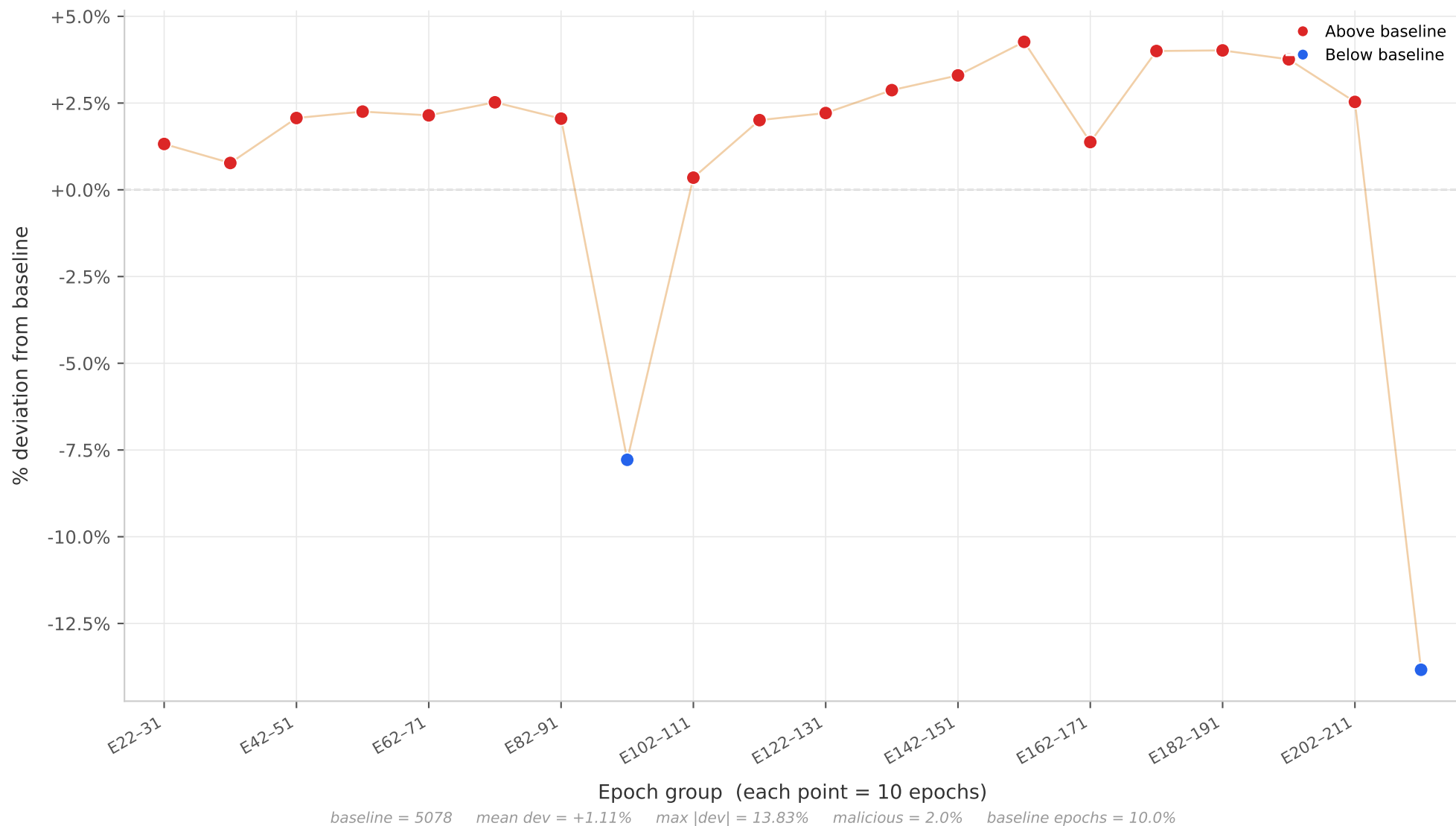
Src / Dst IP Ratio

Flow



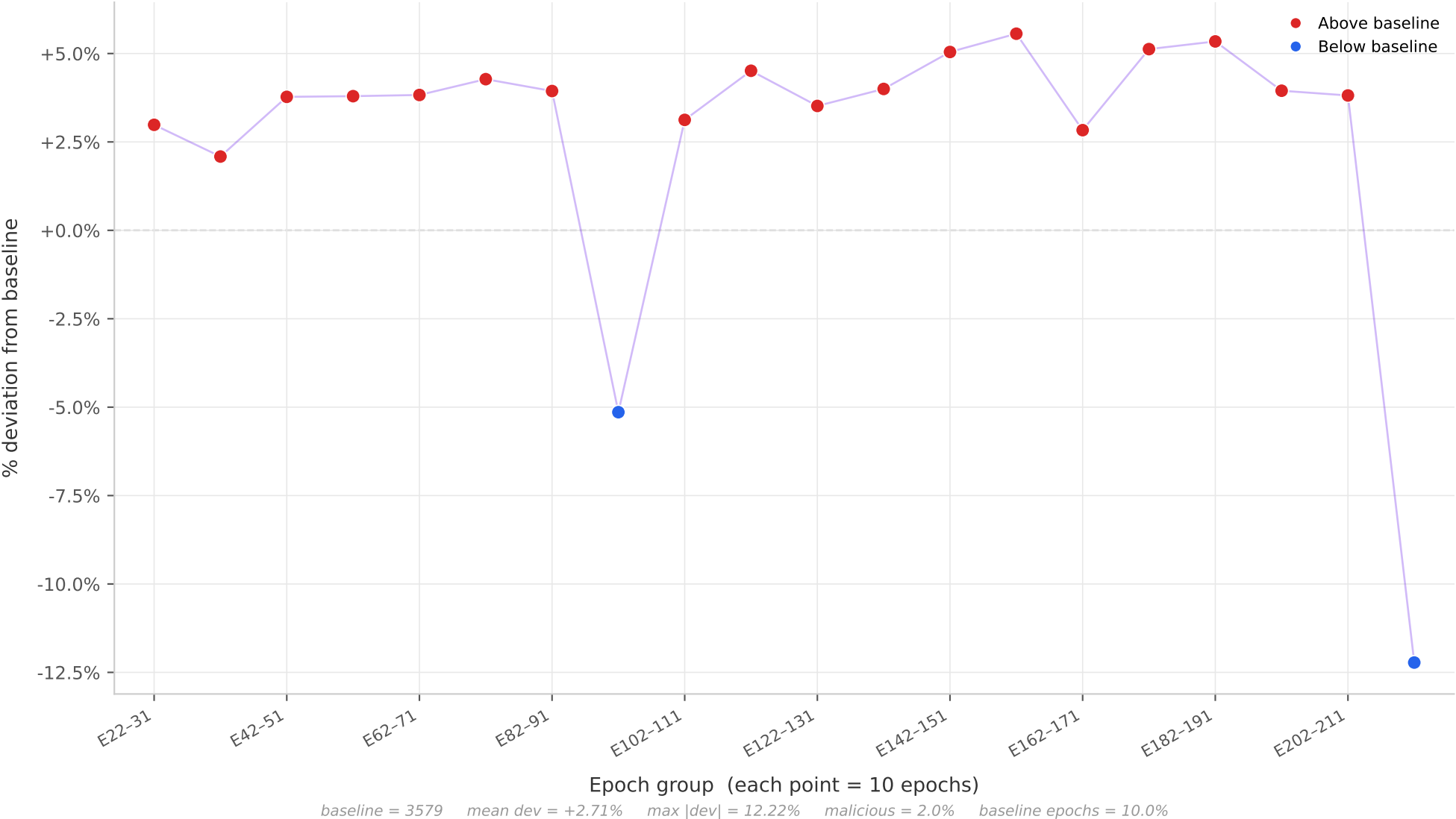
Total Flows

Flow



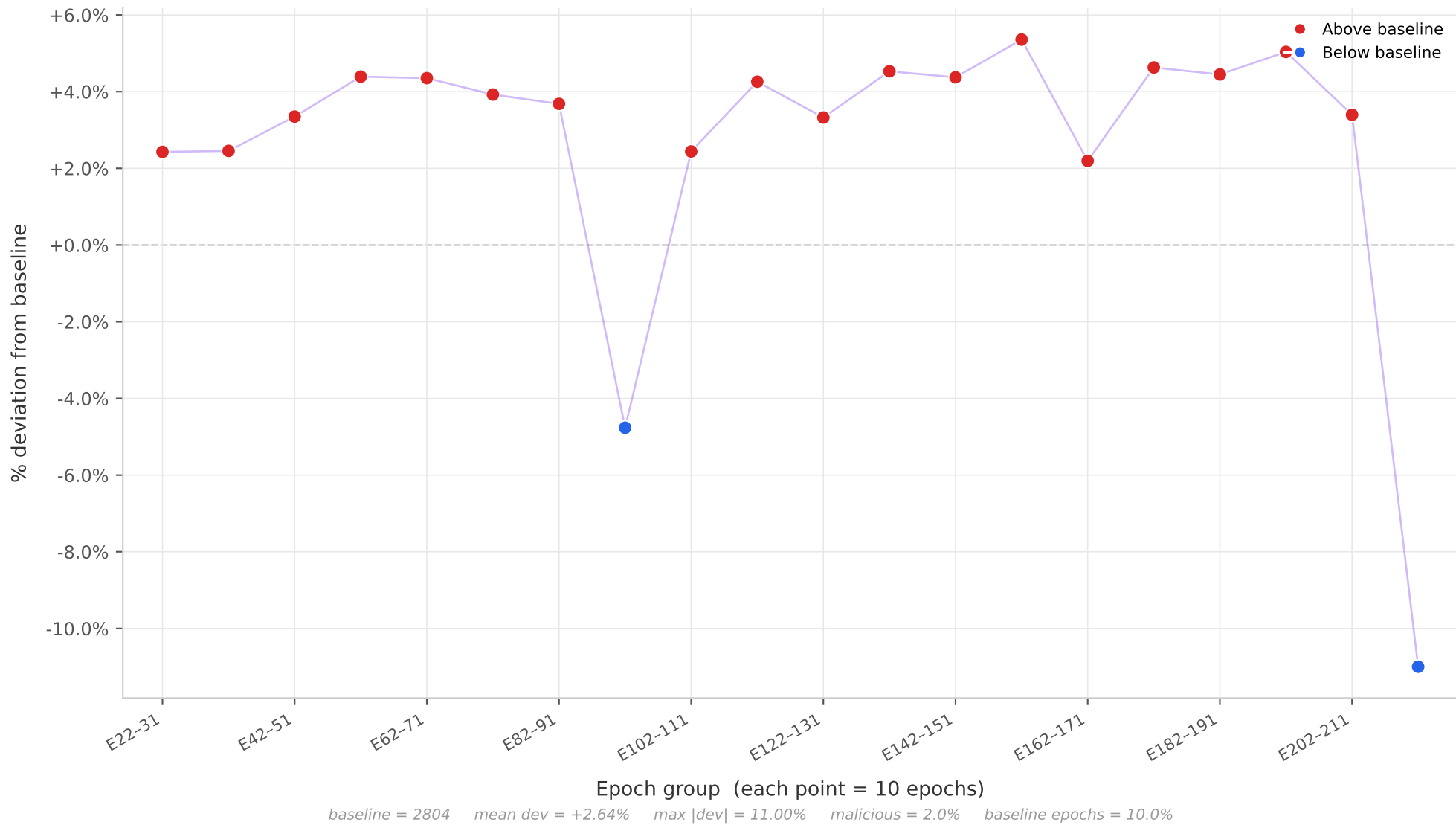
Unique Source IPs

Count



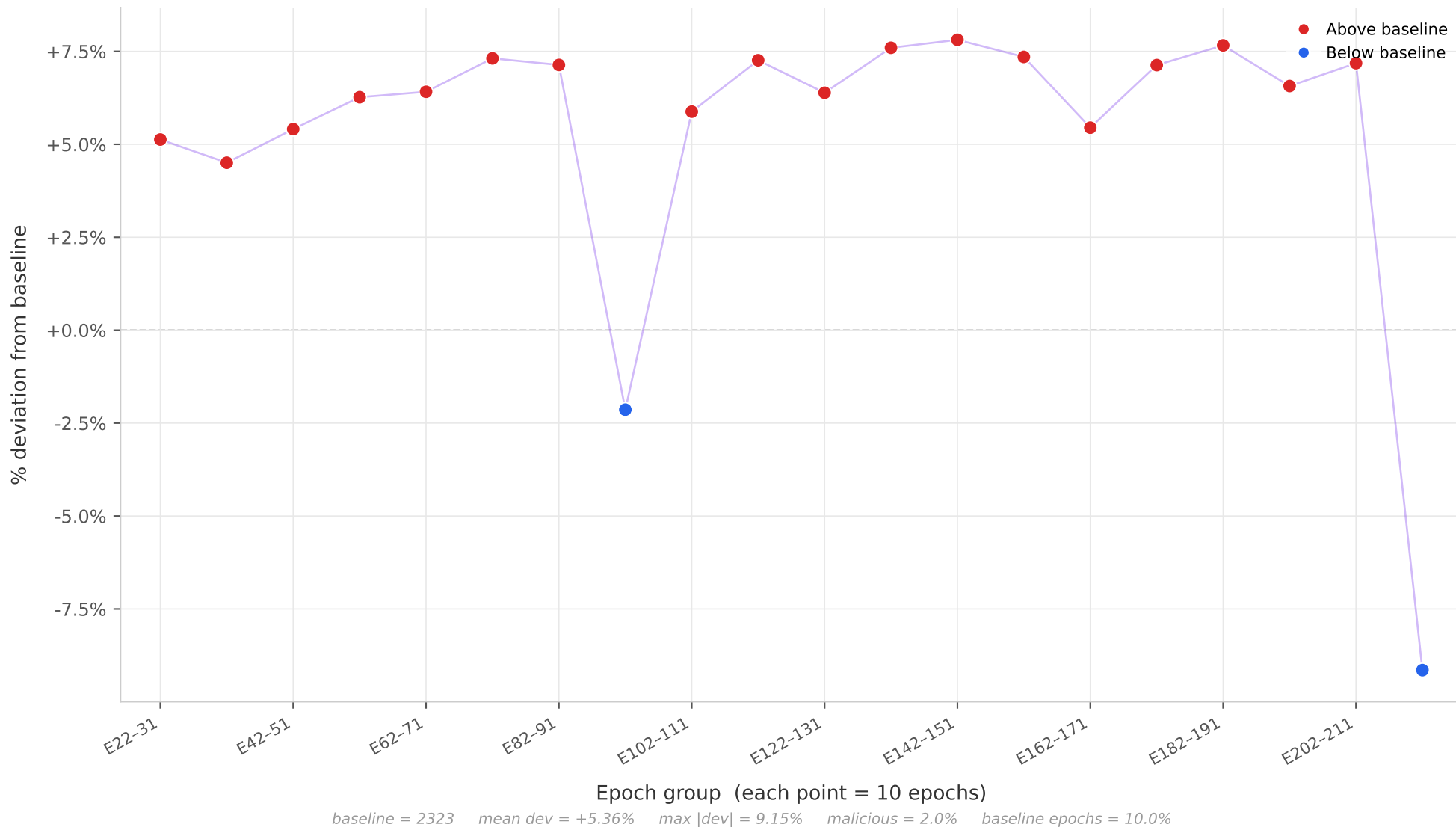
Unique Destination IPs

Count



Unique Source Ports

Count



Unique Destination Ports

Count

